

4.6 TRIG IDENTITIES WORKSHEET #1

① a) $\underline{LS}$ $\underline{RS}$

$$= \sin \theta \quad = \cos \theta \tan \theta$$

$$= \sin \theta \quad = \cancel{\cos \theta} \left(\frac{\sin \theta}{\cancel{\cos \theta}} \right)$$

$$= \sin \theta$$

$LS = RS$

b) $\underline{LS}$ $\underline{RS}$

$$= \csc \theta \quad = \sec \theta \cot \theta$$

$$= \frac{1}{\sin \theta} \quad = \frac{1}{\cos \theta} \left(\frac{1}{\tan \theta} \right)$$

$$= \frac{1}{\cos \theta} \left(\frac{1}{\frac{\sin \theta}{\cos \theta}} \right)$$

$$= \frac{1}{\cos \theta} \left(\frac{\cos \theta}{\sin \theta} \right)$$

$$= \frac{1}{\sin \theta}$$

$LS = RS$

c) $\underline{LS}$ $\underline{RS}$

$$= \cos \theta \quad = \sin \theta \cot \theta$$

$$= \cos \theta \quad = \sin \theta \left(\frac{1}{\tan \theta} \right)$$

$$= \cos \theta \quad = \sin \theta \left(\frac{1}{\frac{\sin \theta}{\cos \theta}} \right)$$

$$= \cos \theta \quad = \cancel{\sin \theta} \left(\frac{\cos \theta}{\cancel{\sin \theta}} \right)$$

$$= \cos \theta$$

$LS = RS$

d) $\underline{LS}$ $\underline{RS}$

$$= \sec \theta \quad = \csc \theta \tan \theta$$

$$= \frac{1}{\cos \theta} \quad = \left(\frac{1}{\sin \theta} \right) \left(\frac{\sin \theta}{\cos \theta} \right)$$

$$= \frac{1}{\cos \theta}$$

$LS = RS$

$$\begin{array}{l|l}
 \textcircled{2} \text{ a) } \quad \underline{LS} & \underline{RS} \\
 = 1 + \csc \theta & = \csc \theta (1 + \sin \theta) \\
 & = \csc \theta + \csc \theta \sin \theta \\
 & = \csc \theta + \left(\frac{1}{\sin \theta} \right) \sin \theta \\
 & = \csc \theta + 1 \\
 \hline
 LS = RS
 \end{array}$$

$$\begin{array}{l|l}
 \text{b) } \quad \underline{LS} & \underline{RS} \\
 = \cot \theta \sin \theta \sec \theta & = 1 \\
 = \left(\frac{\cos \theta}{\sin \theta} \right) \sin \theta \left(\frac{1}{\cos \theta} \right) & \\
 = \frac{\cos \theta}{\cos \theta} & \\
 = 1 & \\
 \hline
 LS = RS
 \end{array}$$

$$\begin{array}{l|l}
 \text{c) } \quad \underline{LS} & \underline{RS} \\
 = \cos \theta (\sec \theta - 1) & = 1 - \cos \theta \\
 = \cos \theta \sec \theta - \cos \theta & \\
 = \cos \theta \left(\frac{1}{\cos \theta} \right) - \cos \theta & \\
 = 1 - \cos \theta & \\
 \hline
 LS = RS
 \end{array}$$

$$\begin{array}{l|l}
 \text{d) } \quad \underline{LS} & \underline{RS} \\
 = 1 + \sin \theta & = \sin \theta (1 + \csc \theta) \\
 & = \sin \theta + \sin \theta \csc \theta \\
 & = \sin \theta + \sin \theta \left(\frac{1}{\sin \theta} \right) \\
 & = \sin \theta + 1 \\
 \hline
 LS = RS
 \end{array}$$

(3)

LS

$$= 1 - \sin^2 \theta$$

$$= \sin^2 \theta + \cos^2 \theta - \sin^2 \theta$$

$$= \cos^2 \theta$$

RS

$$= \sin \theta \cos \theta \cot \theta$$

$$= \sin \theta \cos \theta \left(\frac{\cos \theta}{\sin \theta} \right)$$

$$= \cos^2 \theta$$

$$LS = RS$$

(4)

LS

$$= \csc^2 \theta$$

$$= \frac{1}{\sin^2 \theta}$$

RS

$$= \cot^2 \theta + 1$$

$$= \frac{\cos^2 \theta}{\sin^2 \theta} + \frac{\sin^2 \theta}{\sin^2 \theta}$$

$$= \frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta}$$

$$= \frac{1}{\sin^2 \theta}$$

$$LS = RS$$

(5)

LS

$$= \frac{\cos \theta}{1 + \sin \theta}$$

$$= \frac{\cos \theta}{1 + \sin \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$$

$$= \frac{\cos \theta (1 - \sin \theta)}{1 - \sin^2 \theta}$$

$$= \frac{\cos \theta (1 - \sin \theta)}{\cos^2 \theta}$$

$$= \frac{1 - \sin \theta}{\cos \theta}$$

$$LS = RS$$

(6)

LS

$$= \frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta}$$

$$= \frac{\cos \theta (1 + \sin \theta) + \cos \theta (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}$$

$$= \frac{\cos \theta + \cos \theta \sin \theta + \cos \theta - \cos \theta \sin \theta}{1 - \sin^2 \theta}$$

$$= \frac{2 \cos \theta}{\cos^2 \theta}$$

$$= \frac{2}{\cos \theta}$$

RS

$$= \frac{2}{\cos \theta}$$

$$LS = RS$$

⑦

LS

$$= \csc^2 \theta \cos^2 \theta$$

$$= \frac{1}{\sin^2 \theta} (\cos^2 \theta)$$

$$= \frac{\cos^2 \theta}{\sin^2 \theta}$$

RS

$$= \csc^2 \theta - 1$$

$$= \frac{1}{\sin^2 \theta} - \frac{\sin^2 \theta}{\sin^2 \theta}$$

$$= \frac{1 - \sin^2 \theta}{\sin^2 \theta}$$

$$= \frac{\cos^2 \theta}{\sin^2 \theta}$$

LS = RS

⑧

LS

$$= \tan^2 \theta - \sin^2 \theta$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta} - \frac{\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$$

$$= \frac{\sin^2 \theta (1 - \cos^2 \theta)}{\cos^2 \theta}$$

$$= \tan^2 \theta (1 - \cos^2 \theta)$$

$$= \tan^2 \theta \sin^2 \theta$$

RS

$$= \sin^2 \theta \tan^2 \theta$$

LS = RS